Sai Teja Koppolu

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Professional Summary

Memphis-based Data Science graduate (MS, GPA 3.9) who builds AI agents and Python automations that take manual reporting off people's desks. Built and launched Casebench, an open-source platform where LLM agents brief users and a grader agent scores their work, and automated the hand counting behind **3 leadership tracking streams** at a District Attorney's office. Treats every output as unfinished until its numbers tie back to the source data.

Education

University of Memphis | MS in Data Science | GPA 3.9 Aug 2024 – May 2026
Weil Graduate Fellowship (entrepreneurship-focused, substantial scholarship, leadership training) · Member, NSLS
Jawaharlal Nehru Technological University (JNTU), India | B.Tech in Electronics and Computer Engineering | GPA 3.5

Technical Skills

AI Agents & Automation: LLM Agents, Prompt Engineering, LLM Evaluation, Local LLMs (Ollama), RAG (FAISS)
Python & Spreadsheets: Python (pandas, NumPy, scikit-learn), Excel, pytest, Git/GitHub, Google Colab
Data & Reconciliation: SQL (Window Functions, CTEs), PostgreSQL, DuckDB, Record Linkage, Data-Quality Checks, ETL
Reporting & Monitoring: Power BI (DAX, Power Query), Streamlit, Plotly, KPI Frameworks, Alerting, PowerPoint, Word
Engineering & Modeling: Text Extraction, TypeScript, Next.js, Logistic Regression, Gradient Boosting, Calibration, ROC/AUC

Professional Experience

Data Analyst Intern | Shelby County District Attorney General, *Memphis, TN* 2026 – Present

- Turned a manual counting process into a repeatable Python (pandas), SQL, and Excel workflow, so the **3 leadership tracking streams** (crime trends, task-force performance, crime rates) refresh without hand tallies.
- Made court records without a primary key reconcilable: analyzed how they link and built one standard structure joining **4 record types** (defendants, cases, charges, court events) so each case ties to its source.
- Built a Python and local-LLM extraction step that turns unstructured court-record text into structured, queryable fields while keeping **100%** of sensitive case data on-premises.
- Publish weekly arrest-data KPIs, findings, and recommendations for the task force's Chief Data Officer, including a proposed monitoring report after showing that new and pending cases overlap more with each added case.

Teaching Assistant, Machine Learning & IT Technical Assistant | University of Memphis Nov 2024 – May 2026

- Closed **1,000+ support tickets** in 14 months (about 70 a month) across network, account, hardware, and software issues while completing an MS, and improved the desk's support workflows.
- Built machine learning coursework and assignments with the PhD students on the teaching team, covering **5 model families** in Python (scikit-learn) with cross-validation and ROC analysis.
- Kept the desk's service-record data in PostgreSQL and supported the university's move to new Oracle software and fully digital campus cards; trained and mentored new hires.
- Guided students through ML projects in Google Colab with pandas, NumPy, and Hugging Face, debugging their pipelines and explaining model evaluation results.
- Served as the front-line face of the ITS Service Desk for students, faculty, and staff, handling escalations and helping people in distress while keeping every ticket moving to resolution.

Product Analyst | TCS, *Hyderabad, India* Jul 2023 – Aug 2024

- Defined KPIs and measurement frameworks for product features with Product Managers and Engineers, then tracked user behavior and product metrics in SQL to show the trends and friction points behind feature decisions.
- Delivered the Pantech client engagement inside TCS, preparing the client's raw datasets in Python and Excel and reporting the trends they revealed back to the client.

Projects

Casebench – Open-Source AI Work-Simulation Platform | *TypeScript, Next.js, LLM Agents* github.com/kstc707/casebench

- **Multi-agent platform, built and launched:** LLM-agent coworkers with private knowledge brief the user, who queries data in an in-browser DuckDB SQL sandbox; a grader agent scores deliverables on a **6-criterion rubric**.
- **Ground-truth grading:** the flagship analyst case spans 6 tables and 19,045 sessions; finding a duplicate-event bug cuts the apparent watch-time drop from −21.5% to −**13.8%**, so answers are scored against real data.
- **Owned end to end** over **13 documented build iterations**: benchmarked 10 competing products, redesigned 9 interface surfaces, and released it under the MIT license.

Late-Delivery Prediction & Model Monitoring | *Python, scikit-learn, Power BI* github.com/kstc707/late-delivery-prediction

- **LLM-drafted weekly operations brief:** a local LLM (qwen3:8b) writes the brief, and every number in it is checked against the Power BI tables (10 tables, **32 DAX measures**) before release.
- **Source-tied data prep** on 180,519 line items rolled up to **65,752 orders**: excluded 2,855 cancelled orders with wrong late flags and mapped 164 country names to ISO codes to join holiday calendars.
- **Automated monitoring** over 13 months (calibration gap never above 1.3 points, 0 model alerts); a simulated process change trips both alerts in its first month, with alert rules covered by **18 unit tests**.

Automated Data-Quality & Anomaly-Detection Pipeline | *Python, pandas, NumPy* github.com/kstc707/data-quality-pipeline

- **Reusable quality gate** of 8 automated checks (schema, nulls, duplicates, Z-score/IQR outliers, business rules, dates) that runs before any number reaches a report.
- **Reconciliation results:** on an 8,160-row seeded test set it flagged the **15.4%** of rows failing a check, including **262 reconciliation failures** and 149 duplicates, and now gates the late-delivery data.